less numerous in the gene pool. The task of an efficient egg-layer is one of predicting what is going to be the optimum clutch size for her, as a selfish individual, in the coming breeding season. You will remember from Chapter 4 the special sense in which we are using the word prediction. Now how can a female bird predict her optimum clutch size? What variables should influence her prediction? It may be that many species make a fixed prediction, which does not change from year to year. Thus on average the optimum clutch size for a gannet is one. It is possible that in particular bumper years for fish the true optimum for an individual might temporarily rise to two eggs. If there is no way for gannets to know in advance whether a particular year is going to be a bumper one, we cannot expect individual females to take the risk of wasting their resources on two eggs, when this would damage their reproductive success in an average year.

But there may be other species, perhaps starlings, in which it is in principle possible to predict in winter whether the following spring is going to yield a good crop of some particular food resource. Country people have numerous old sayings suggesting that such clues as the abundance of holly berries may be good predictors of the weather in the coming spring. Whether any particular old wives' tale is accurate or not, it remains logically possible that there are such clues, and that a good prophet could in theory adjust her clutch size from year to year to her own advantage. Holly berries may be reliable predictors or they may not but, as in the case of the mice, it does seem quite likely that population density would be a good predictor. A female starling can in principle know that, when she comes to feed her babies in the coming spring, she will be competing for food with rivals of the same species. If she can somehow estimate the local density of her own species in winter, this could provide her with a powerful means of predicting how difficult it is going to be to get food for babies next spring. If she found the winter population to be particularly high, her prudent policy, from her own selfish point of view, might well be to lay relatively few eggs: her estimate of her own optimum clutch size would have been reduced.

Now the moment it becomes true that individuals are reducing their clutch size on the basis of their estimate of population density, it will immediately be to the advantage of each selfish individual to pretend to rivals that the population is large, whether it really is or not. If starlings are estimating population size by the volume of noise in a winter roost, it